Smart Energy Management Using Forecast-Driven Generation Optimization

1. Problem Statement: Electricity demand changes every hour due to factors such as consumer usage patterns, weather conditions, and time of day, while power generation must be scheduled in advance. Most existing power systems still depend on reactive control, which responds only after demand increases, leading to inefficiencies and higher operating costs. The growing use of renewable energy further increases uncertainty in supply and demand balance. This results in higher peak loads, increased stress on grid infrastructure, and energy wastage. Therefore, there is a need for predictive load scheduling to support generation planning and improve overall grid reliability.

2. Dataset: Our project uses the Open STEF Liander 2024 Energy Forecasting Benchmark dataset. The dataset is designed for short-term energy forecasting under real operational grid conditions. It consists of electrical load measurements recorded at a 15-minute resolution across 55 grid locations in the Netherlands over the full year of 2024. In addition to historical load data, the dataset includes relevant exogenous variables such as weather conditions and day-ahead electricity prices. These features capture realistic demand variability, making the dataset well suited for evaluating forecast-driven generation planning and operational decision-making in power systems.

3. Methodology / Strategy: A hybrid CNN–LSTM framework is employed for short-term electricity demand forecasting using multivariate time-series inputs derived from historical load and exogenous variables such as weather data. The data is structured using a sliding-window approach, where CNN layers learn local temporal patterns including short-term fluctuations, while LSTM layers capture longer-term dependencies across time. The model generates near-future load forecasts, which are then provided to a rule-based and decision tree–based planning module. This control layer applies predefined operational thresholds to proactively schedule generation capacity, aiming to reduce overgeneration, mitigate peak demand stress, and improve system stability compared to reactive planning (No-forecast-based scheduling).

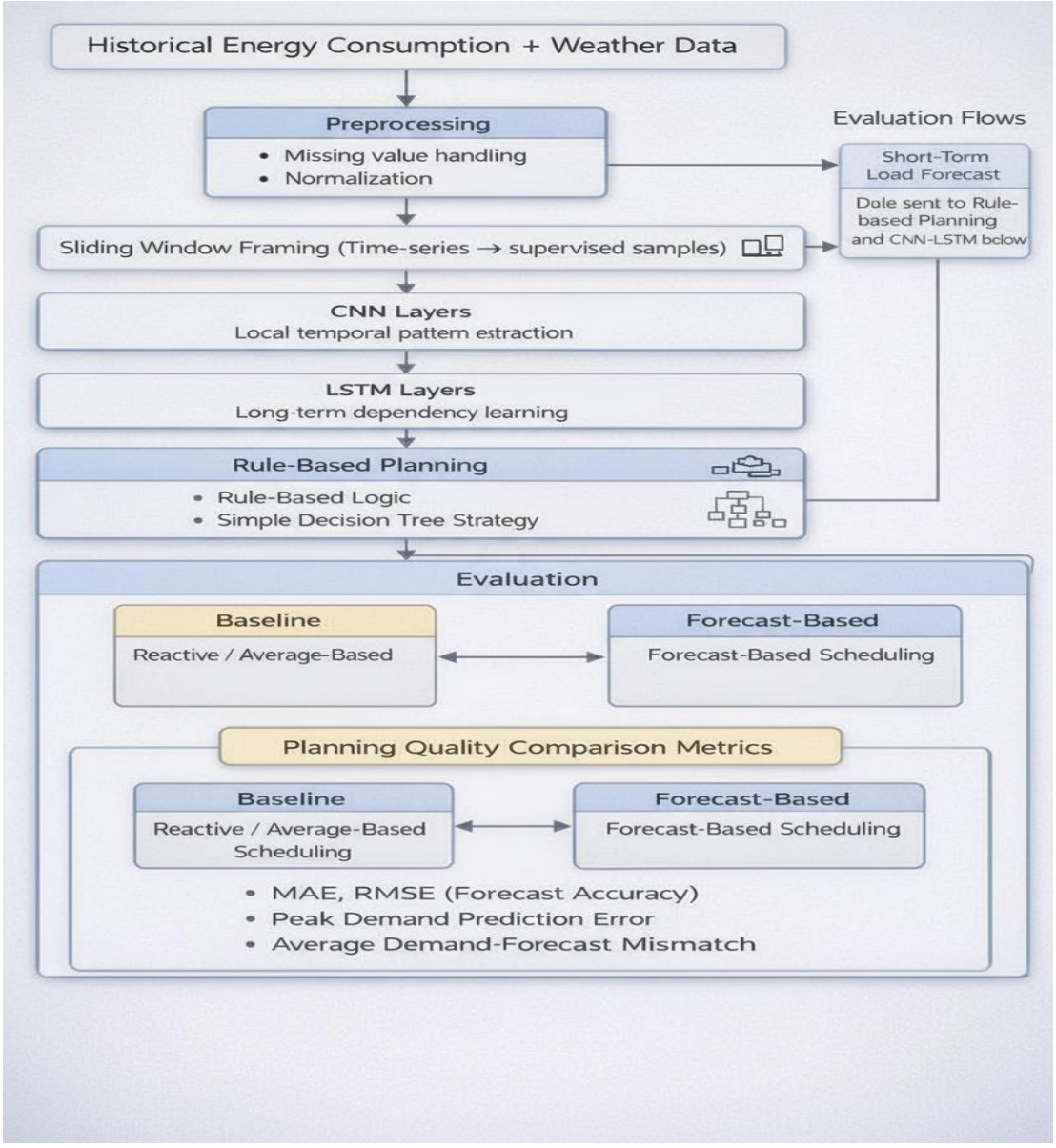
4. Relevance: This project targets short-term electricity demand forecasting for efficient and cost-effective grid operation. A CNN–LSTM hybrid model is selected over classical statistical methods and traditional machine learning models due to its ability to automatically learn complex, nonlinear load patterns and long-term temporal dependencies. Multi-feature inputs combining load and weather data provide a more realistic and generalized forecasting setup than simple timestamp–load models. Evaluation compares forecast-driven scheduling with a reactive baseline using forecasting error and demand-mismatch metric. To ensure transparency and interpretability, rule-based/decision tree–based optimization is used instead of reinforcement learning.

5. References:

- Research Paper 1

- Research Paper 2
- Research Paper 3
- Research Paper 4
- Dataset

Architecture Diagram

- 

Model Architecture and Validation Strategy:

The system architecture consists of data preprocessing and sliding-window framing of multivariate load and weather inputs, followed by CNN layers for local temporal feature extraction and LSTM layers for modeling long-term dependencies. The trained CNN–LSTM model generates short-term load forecasts, which are passed to a rule-based /decision tree–based planning module for proactive generation scheduling. System performance is evaluated by comparing forecast-driven planning against a reactive baseline using standard forecasting accuracy metrics (MAE and RMSE). Planning effectiveness is further assessed using mismatch-based proxy metrics, including peak demand prediction error and average demand–forecast deviation.